DGMS 2019 • SECTION G • QUESTION 31

Global Seismology & Tectonics: Earthquake Focal Depths & The Wadati-Benioff Zone

Directorate General of Mines Safety (DGMS) Junior Scientific Officer Examination 2019 | Seismotectonics & Plate Margins Guide

Question Statement

Q.31. The deep-focused earthquakes commonly originate along:

a (a) Mid-oceanic ridge

b (b) Continental rift

c (c) Transform fault

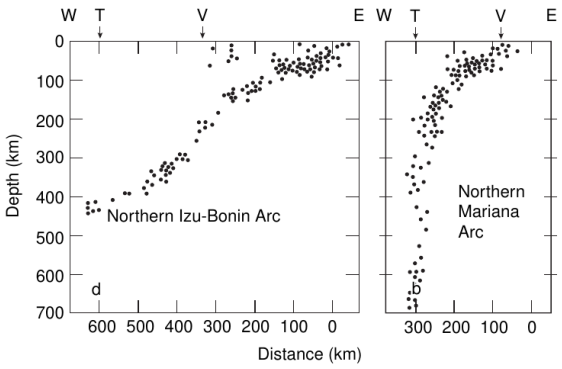
d **(d) Benioff zone**

Correct Answer: (d) Benioff zone (Wadati-Benioff subduction zone down to ~670 km)

Official Textbook Figure (Van der Pluijm & Marshak, Earth Structure)



(a)



(b)

core-mantle boundary (Figure 17.9). If this image is correct, then the base of the Wadati-Benioff zone does not mark the base of subducted slabs, but merely the depth at which earthquakes no longer occur because the fracturing and/or phase changes that produce seismic energy in slabs can no longer take place.

17.2.2 The Trench

Trenches are linear or curvilinear troughs that mark the boundary, at the Earth's surface, between the downgoing slab and the accretionary prism of the overriding plate (Figure 17.4). Trenches exist because the subducted portion of the downgoing slab pulls the slab downwards to a

FIGURE 17.6 (a) Map of the western Pacific, showing trenches (heavy lines), and volcanoes (black dots) related to subduction in the western Pacific. The depth to Wadati-Benioff zones is shown by contour lines. The contours are given in multiples of 50 km (e.g., "2" means $2 \times 50 = 100$ km depth below the surface.) (b) Cross section showing earthquake foci defining the moderately dipping Wadati-Benioff zone of the northern Izu-Bonin arc. (c) Cross section showing earthquake foci defining the steeply dipping Wadati-Benioff zone of the northern Mariana Arc. T = location of trench; V = location of volcanic arc; the distance between T and V is the arc-trench gap.

Figure 1: B. A. van der Pluijm & S. Marshak, Chapter 17, Figure 17.6. Cross-sections showing earthquake foci defining the inclined **Wadati-Benioff zone** in the Western Pacific (Izu-Bonin and Mariana Arcs). Earthquakes penetrate continuously from shallow depths (< 70 km) near the trench through intermediate depths (70–300 km) down to **deep-focus depths (300 to ~670 km)** strictly within the cold, descending subducted lithospheric slab.

In-Depth Option Evaluation & Seismotectonic Architecture

(d) Benioff zone (Wadati-Benioff Zone) — [CORRECT]

Seismotectonic Mechanism: Named after seismologists Kiyoo Wadati and Hugo Benioff, the Wadati-Benioff zone is a dipping planar zone of seismicity tracing the trajectory of cold, dense oceanic lithosphere subducting into the warm mantle asthenosphere.

Focal Depth Distribution: Because ordinary mantle at depths > 50–70 km is too hot and ductile for brittle frictional sliding, earthquakes can only occur where cold,

rigid rock is transported downwards. The subducting slab remains cold and brittle down to **670–700 km** (the upper mantle transition zone), enabling **deep-focus earthquakes (300–670 km)** via dehydration embrittlement and transformational faulting (metastable olivine → spinel/ringwoodite phase changes). **Subduction zones are the ONLY tectonic environments on Earth where deep-focus earthquakes occur!**

(a) Mid-oceanic ridge — [INCORRECT • SHALLOW ONLY]

Mid-ocean ridges are divergent plate boundaries characterized by high heat flow and thin, newly formed lithosphere. Brittle failure is strictly limited to the uppermost **10–15 km** (shallow-focus earthquakes only) along extensional rift grabens.

(b) Continental rift — [INCORRECT • SHALLOW ONLY]

Continental rifts (e.g., East African Rift, Baikal Rift) involve crustal thinning and elevated geothermal gradients. Earthquakes are restricted to the brittle upper crust at depths **< 20–30 km** (shallow-focus).

(c) Transform fault — [INCORRECT • SHALLOW ONLY]

Transform faults (e.g., San Andreas Fault, Alpine Fault, oceanic fracture zones) represent strike-slip motion between plates. Frictional locking and rupture occur exclusively in the upper seismogenic crust, rarely exceeding **15–20 km** depth.

Global Earthquake Focal Depth Classification & Plate Margins

Classification	Focal Depth (h)	Typical Tectonic Environments	Dominant Rupture Mechanism
Shallow-Focus	**0 – 70 km**	Mid-ocean ridges, continental rifts, transform faults, collision zones, subduction trenches	Classic brittle frictional sliding & cataclasis (Byerlee's Law)
Intermediate-Focus	**70 – 300 km**	**Subduction zones exclusively** (Wadati-Benioff zone)	Dehydration embrittlement of serpentinite/hydrous minerals
Deep-Focus	**300 – 670 km**	**Wadati-Benioff zones exclusively (Subduction slabs)**	**Transformational faulting (metastable olivine → spinel/ wadsleyite transition)**

Direct Textbook Quotations & Citations

B. A. van der Pluijm & S. Marshak (2004), *Earth Structure: An Introduction to Structural Geology and Tectonics, 2nd ed., p. 217 & 434:*

"Earthquakes in subduction zones define an inclined belt, called the **Wadati-Benioff zone***, that extends from the trench down to depths as great as* **670 km***... In fact, it is the distribution of earthquakes in the Wadati-Benioff zone that outlines the geometry of subducting slabs. Deep-focus earthquakes occur only where cold, rigid slabs penetrate into the mantle."*

— Van der Pluijm & Marshak, p. 217 & 434

William Lowrie (2007), *Fundamentals of Geophysics, 2nd ed., Cambridge University Press, Chapter 3:*

"Deep-focus earthquakes (depths > 300 km) are unknown along mid-ocean ridges or transform faults. They occur solely along the Wadati-Benioff zones associated with subduction at convergent plate boundaries, terminating abruptly at the 670 km seismic discontinuity."

— William Lowrie, Fundamentals of Geophysics

Exam Traps & Rapid Elimination Strategy

Rule of Thumb for Earthquake Depths:

1. **Mid-Ocean Ridges & Transform Faults:** NEVER have deep earthquakes! Always shallow (< 20 km).
2. **Continental Rifts:** Always shallow (< 30 km).
3. **Deep-Focus Earthquakes (> 300 km):** Found **ONLY in Wadati-Benioff zones** of subduction zones!